

Retail Sales Analytics Database

Database Design

This project was developed to manage and analyze the sales activities of an online retail business. The database stores information about customers, products, orders, payments, and deliveries in an organized manner.

A total of six tables were created:

- Customers – stores customer details such as name, city, gender, and membership type.
- Products – stores product information, category, and price.
- Orders – stores details of orders placed by customers.
- Order_Items – stores the products included in each order along with quantity.
- Payments – stores payment method, amount, and payment status.
- Deliveries – stores delivery partner information and delivery status.

Primary keys were used in all tables to uniquely identify each record and avoid duplicate entries.

Table Relationships

The tables are connected using foreign keys to maintain proper relationships between the data.

The relationships are:

- Orders.customer_id → Customers.customer_id
- Order_Items.order_id → Orders.order_id
- Order_Items.product_id → Products.product_id
- Payments.order_id → Orders.order_id
- Deliveries.order_id → Orders.order_id

These relationships help ensure that orders are linked to valid customers, products, payments, and deliveries. They also improve data accuracy and consistency within the database.

Key Insights from Reports

Several SQL queries were executed to analyze the data and generate useful business insights.

Some important observations include:

- Customers were spread across different cities, with some cities having a higher customer count than others.
- Electronics and Fashion products contributed significantly to overall sales.
- A few customers placed multiple orders, showing repeat purchasing behavior.
- Successful payments generated the majority of the revenue.
- Revenue analysis helped identify valuable customers and high-performing product categories.
- Delivery reports helped track pending and completed deliveries.
- Payment status reports made it easy to identify successful and failed transactions.
- Product sales reports highlighted the most frequently purchased products.

Conclusion

The Retail Sales Analytics Database successfully demonstrates how a retail business can organize and manage its operational data using MySQL. Through the use of joins, aggregate functions, subqueries, and data quality checks, meaningful reports were generated to support business analysis and decision-making. This project helped in understanding database design concepts, table relationships, and SQL-based reporting techniques in a practical business scenario.

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